



Data Analytics with Pandas: An Introduction

Data science has emerged as an essential skill today. Analysing Big Data to generate insights that help businesses profit more is one of the key skills required to be a data scientist. The Pandas library is an open source data analytics tool in Python. It is a must-learn for every aspiring data scientist.



In an article, the *Harvard Business Review* rated 'data scientist' as the sexiest job to have in the 21st century. This is due to multiple reasons, the primary one being the abundance of Big Data available today, and the potential for us to draw key insights from it that could help companies make more profits. Another important development has been the advent of Python as the single most in-demand programming language for data analytics. This is due to the vast variety of libraries available in Python to clean, analyse and visualise data.

In this article, I will introduce you to the Pandas library, an open source

Python based data analytics tool that helps us organise and structure data for easier and intuitive analysis of data.

What is Pandas?

From its initial release in 2008, Pandas has emerged as the most popular open source tool for organising, structuring and manipulating data. Its latest version (1.0.3) was released in March 2020. The library is written in Python, Cython and C. It has a huge and active open source community that supplements its development, and has big sponsors such as Anaconda and the Chan Zuckerberg Initiative, to name a few.

Using Pandas, you can load csv, xlsx, zip, txt, json, xml, html, images, hierarchical data, pdf, docx, mp3, mp4 or SQL data and visualise it in the form of structured columns and rows. Further, you can manipulate this data easily to add, edit or delete rows and columns.

Installing Pandas

Installing Pandas is as straightforward as installing any other library in Python. Just open your terminal (make sure you have Pip installed) and run the following piece of code:

```
pip install pandas
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